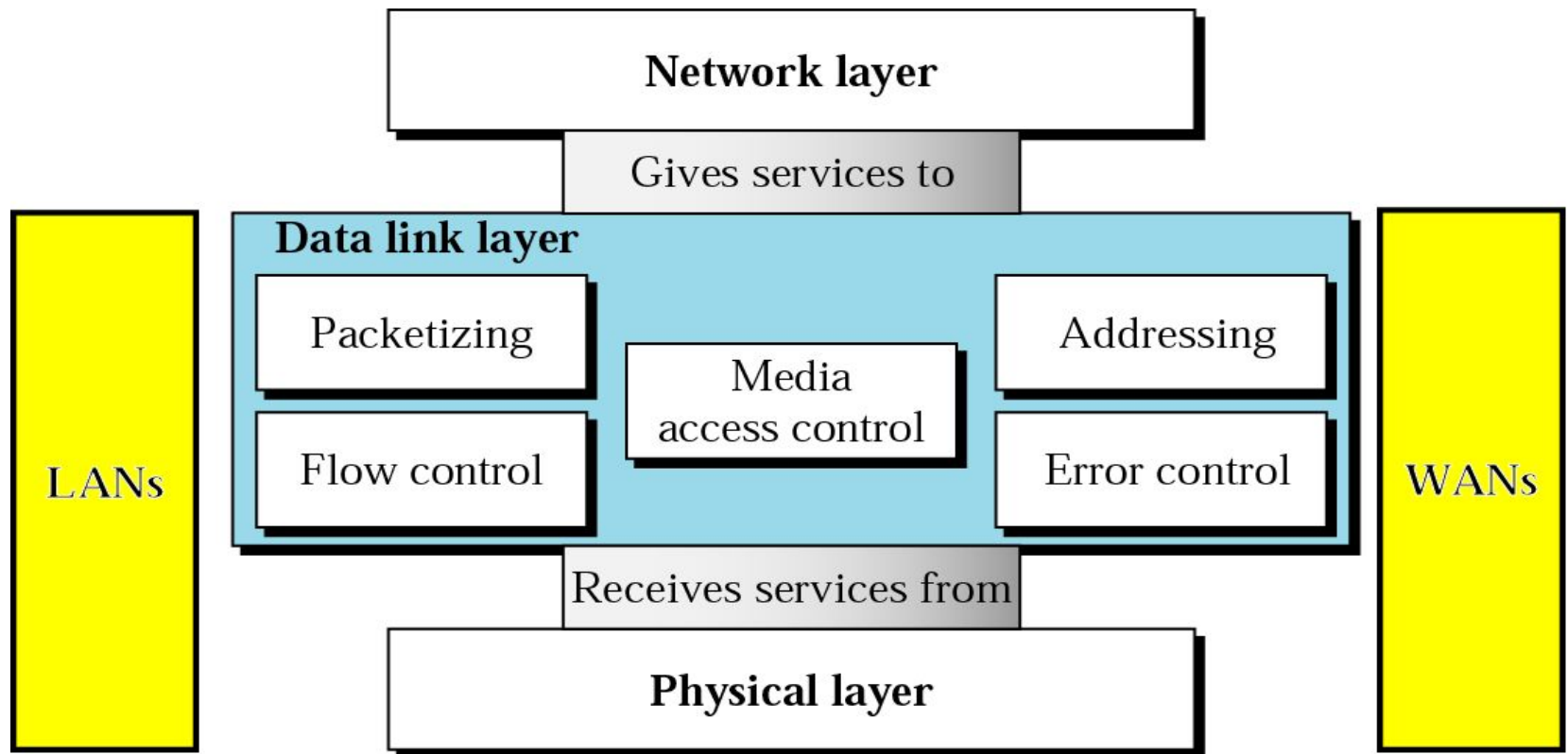


Computer Communication & Networks

Lecture 12

Datalink Layer: Error Correction

Data Link Layer



Data Link Layer Topics to Cover

Error Detection and Correction

Data Link Control and Protocols

Multiple Access

Local Area Networks

Wireless LANs

Error Correction

1. By retransmission
 - ❑ flow and error control protocols
2. Forward Error Correction (FEC)
 - ❑ require more redundancy bits
 - ❑ should locate the invalid bit or bits
 - ❑ n -bit code word contains m data bits + r redundancy bits
$$n = m + r$$
 - ❑ $m + r + 1$ bits discoverable by r bits
 - ❑ $2^r \geq m + r + 1$

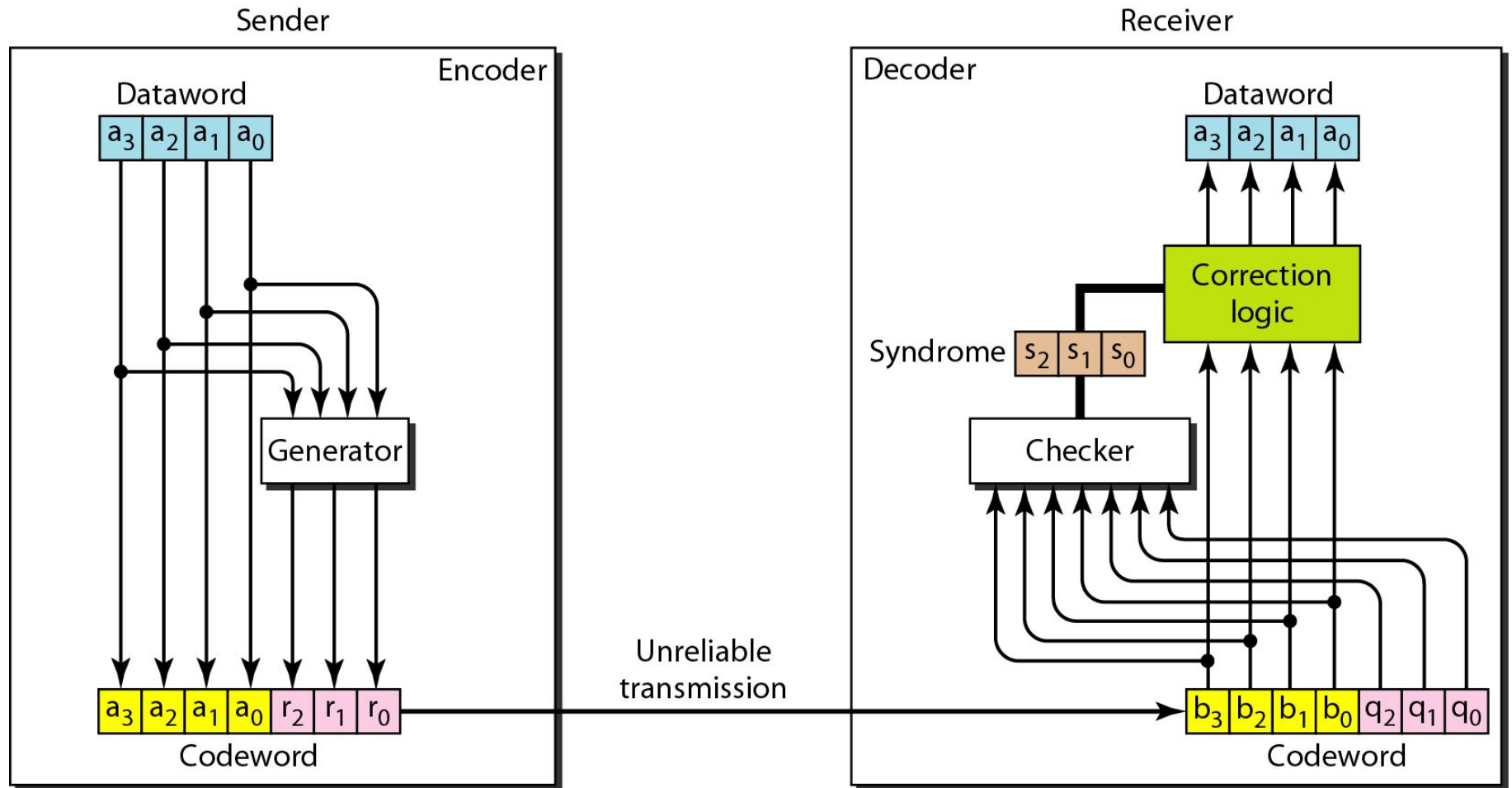
Data and redundancy bits

Number of data bits m	Number of redundancy bits r	Total bits $m + r$
1	2	3
2	3	5
3	3	6
4	3	7
5	4	9
6	4	10
7	4	11

Hamming Code

- Hamming codes provide for FEC using a “Block Parity”
 - i.e, instead of one parity bit send a block of parity bits
 - Allows correction of single bit errors
 - This is accomplished by using more than one parity bit
 - Each computed on different combination of bits in the data
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Hamming code (Contd.)



Positions of Redundancy Bits

11	10	9	8	7	6	5	4	3	2	1
d	d	d	r_8	d	d	d	r_4	d	r_2	r_1

Redundancy Bits Calculation

r_1 will take care of these bits.

11		9		7		5		3		1
d	d	d	r_8	d	d	d	r_4	d	r_2	r_1

r_2 will take care of these bits.

11	10			7	6			3	2	
d	d	d	r_8	d	d	d	r_4	d	r_2	r_1

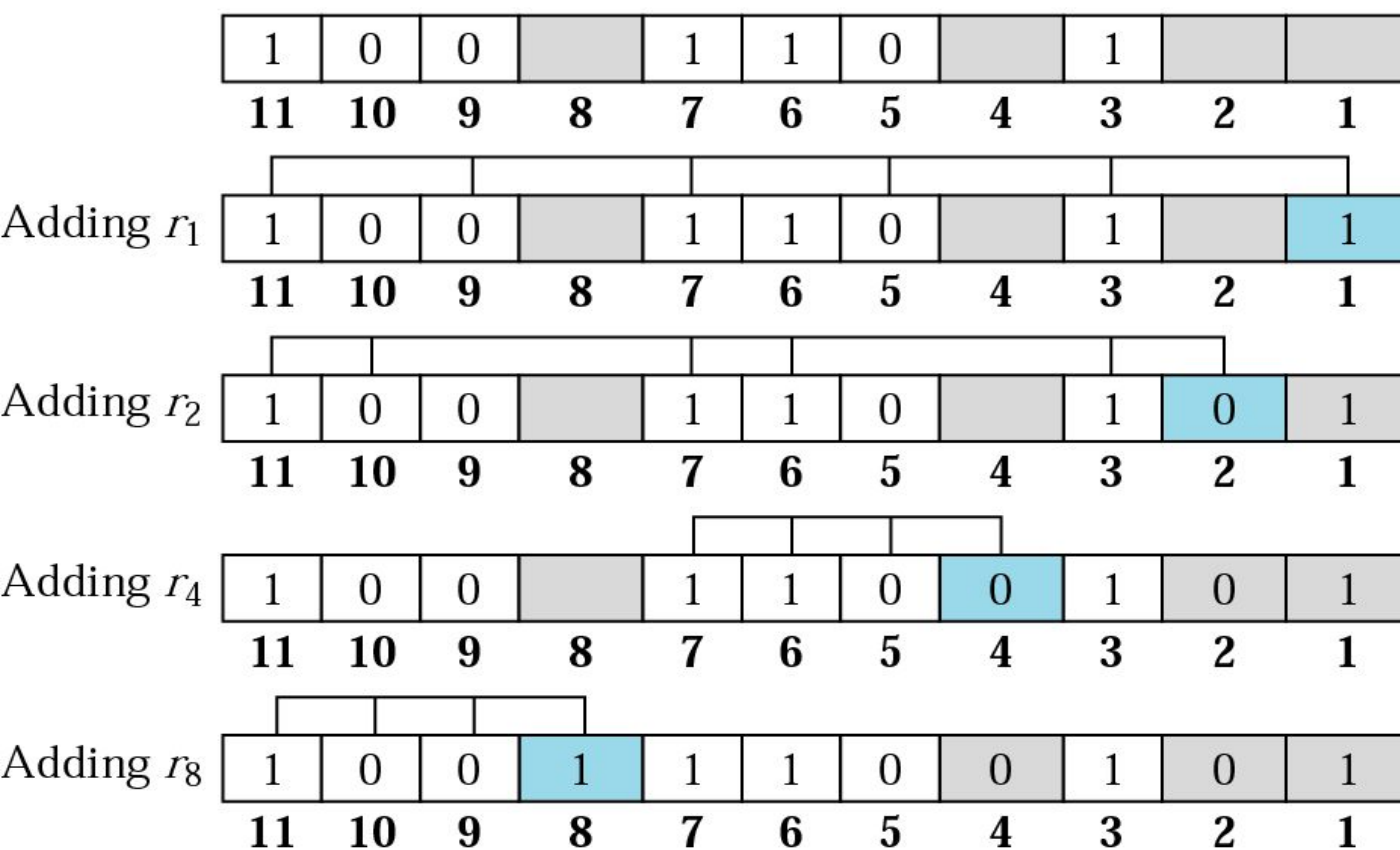
r_4 will take care of these bits.

				7	6	5	4			
d	d	d	r_8	d	d	d	r_4	d	r_2	r_1

r_8 will take care of these bits.

11	10	9	8							
d	d	d	r_8	d	d	d	r_4	d	r_2	r_1

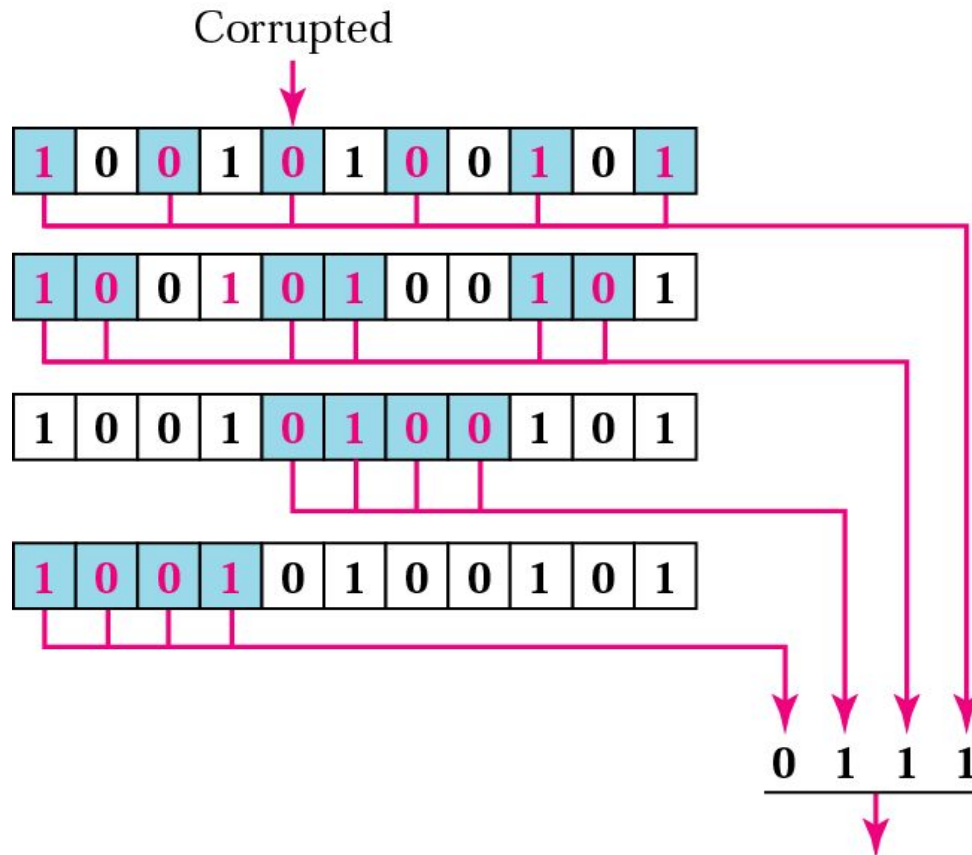
Example



Data:
1001101

Code:
10011100101

Error Correction using Hamming Code



The bit in position 7 is in error. 7

Burst Error Correction

Error → 1111?000011

Error → 1010?011111

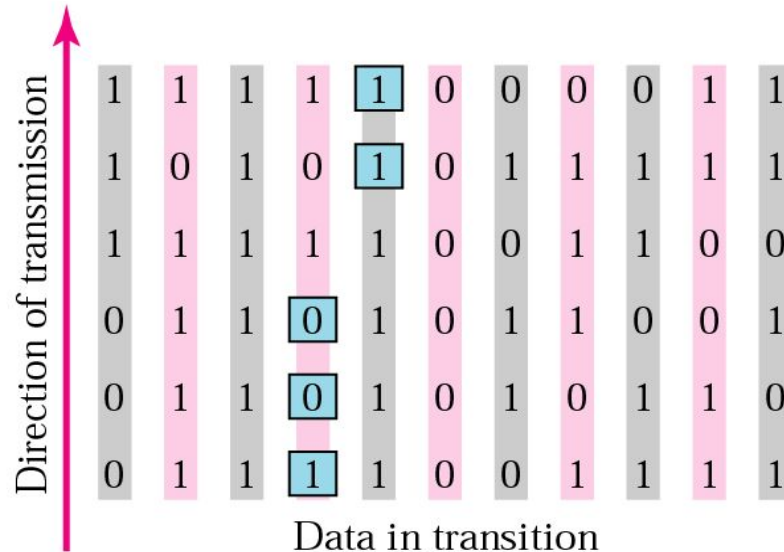
11111001100

Error → 011?1011001

Error → 011?1010110

Error → 011?1001111

Received data



11111000011

10101011111

11111001100

01101011001

01101010110

01111001111

Data before being sent

Hamming Distance

- The Hamming distance between two words is the number of differences between corresponding bits.

Let us find the Hamming distance between two pairs of words.

1. The Hamming distance $d(000, 011)$ is 2 because

$$000 \oplus 011 \text{ is } 011 \text{ (two 1s)}$$

2. The Hamming distance $d(10101, 11110)$ is 3 because

$$10101 \oplus 11110 \text{ is } 01011 \text{ (three 1s)}$$

Note

To guarantee the detection of up to s errors in all cases, the minimum Hamming distance in a block code must be $d_{\min} = s + 1$.

Example

Consider the codeword set presented :

000000 110100 011010 101110 101001 011101 110011 000111

Suppose that the codeword 110011 was transmitted and that the two leftmost digits were declared by the receiver to be erasures. Verify that the received flawed sequence xx0011 can be corrected.

Solution

Since $d_{\min} = \rho + 1 = 3$, the code can correct as many as $\rho = 2$ erasures. This is easily verified above by comparing the rightmost four digits of xx0011 with each of the allowable codewords. The codeword that was actually transmitted is closest in Hamming distance to the flawed sequence.

Note

To guarantee correction of up to t errors in all cases, the minimum Hamming distance in a block code must be $d_{\min} = 2t + 1$.

Example

A code scheme has a Hamming distance $d_{\min} = 4$. What is the error detection and correction capability of this scheme?

Solution

*This code guarantees the detection of up to **three** errors ($s = 3$), but it can correct up to **one** error. In other words, if this code is used for error correction, part of its capability is wasted. Error correction codes need to have an odd minimum distance (3, 5, 7, ...).*

Readings

- Chapter 10 (B.A Forouzan)
 - Section 10.2,10.5
 - (Cover only those contents which are related to topics covered in class)
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